

Halloween Crossword

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ONE ACROSS

At the annual Monster Ball, 6 distinct monsters hang their custom cloaks on coat pegs in the foyer. At midnight, a mischief spell randomly redistributes the cloaks so that exactly two of the monsters receive their own original cloaks back, while the other four monsters each end up with someone else's cloak. In how many distinct ways can the cloaks be redistributed under these conditions?

ONE DOWN

At a Halloween party, 16 identical candy eyeballs are divided among three distinct monsters: a vampire, a werewolf, and a mummy. Each monster must receive at least one candy eyeball, but the vampire's tiny treat bag can hold **at most** 12 candy eyeballs. In how many different ways can the candy eyeballs be distributed?

TWO DOWN

Three enchanted jack-o'-lanterns have magical power levels a , b , and c , which are the three roots of

$$t^3 - 30t^2 + 291t - 910 = 0.$$

A witch needs the value of $a^2 + b^2 + c^2$ to complete her spell. What is this value?

THREE DOWN

A gravekeeper has fewer than 693 glowing skulls in a haunted crypt. When he arranges them in groups of 7, there are 6 skulls left over. When he arranges them in groups of 9, there are 7 skulls left over, and when he arranges them in groups of 11, there are 10 skulls left over. How many glowing skulls are in the crypt?

FOUR ACROSS

Inside a spooky haunted clock tower, a cursed grandfather clock strikes 17^{103} times at the stroke of midnight. To figure out the clue for the crossword, find the remainder when 17^{103} is divided by 100, representing the last two digits of the total number of strikes.

FIVE ACROSS

Three friendly ghouls, Gus, Hank, and Barnaby, are tallying up their Halloween candy counts, denoted as x , y , and z . They notice three curious algebraic facts about their loot: the sum of their candies is $x + y + z = 12$, the sum of the pairwise products is $xy + yz + zx = 44$, and the product of all three candy counts is $xyz = 48$. What is the sum of the cubes of their individual candy counts, $x^3 + y^3 + z^3$?